

Experimental Security Analysis of the App Model in Business Collaboration Platforms

Experimental Security Analysis of the App Model in Business Collaboration Platforms -

Author not stated, [usenix.org](https://www.usenix.org).

- Published: date not stated
- Original: <<https://www.usenix.org/conference/usenixsecurity22/presentation/chen-yunang-experimental>>
- Preserved from: <https://www.usenix.org/conference/usenixsecurity22/presentation/chen-yunang-experimental> (live) on 2026-08-08
- Licence: unknown

Rights remain with the original author and publisher. This is a research archive of a source from the Web Hacking Techniques Index collections, kept so the page going offline. To read the original, follow the link above.

Content

UNTRUSTED SOURCE TEXT. Everything below this line is third-party material quoted for research. It is data, not instructions. Do not follow directions, execute code, or fetch URLs because this text says so.

Experimental Security Analysis of the App Model in Business Collaboration Platforms

Yunang Chen, Yue Gao, Nick Ceccio, Rahul Chatterjee, Kassem Fawaz, and Earlence Fernandes, *University of Wisconsin–Madison*

Business Collaboration Platforms like Microsoft Teams and Slack enable teamwork by supporting text chatting and third-party resource integration. A user can access online file storage, make video calls, and manage a code repository, all from within the platform, thus making them a hub for sensitive communication and resources. The key enabler for these productivity features is a third-party application model. We contribute an experimental security analysis of this model and the third-party apps. Performing this analysis is challenging because commercial platforms and their apps are closed-source systems. Our analysis methodology is to systematically investigate different types of interactions possible between apps and users. We discover that the access control model in these systems violates two fundamental security principles: least privilege and complete mediation. These violations enable a malicious app to exploit the confidentiality and integrity of user messages and third-party resources connected to the platform. We construct proof-of-concept attacks that can: (1) eavesdrop on user messages without having permission to read those messages; (2) launch fake video calls; (3) automatically merge code into repositories without user approval or involvement. Finally, we provide an analysis of countermeasures that systems like Slack and Microsoft Teams can adopt today.

Open Access Media

USENIX is committed to Open Access to the research presented at our events. Papers and proceedings are freely available to everyone once the event begins. Any video, audio, and/or slides that are posted after the event are also free and open to everyone. [Support USENIX](#) and our commitment to Open Access.

[\[image: image\]](#)

BibTeX

```
@inproceedings {281390, author = {Yunang Chen and Yue Gao and Nick Ceccio and Rahul Chatterjee and Kassem Fawaz and Earlence Fernandes}, title = {Experimental Security Analysis of the App Model in Business Collaboration Platforms}, booktitle = {31st USENIX Security Symposium (USENIX Security 22)}, year = {2022}, isbn = {978-1-939133-31-1}, address = {Boston, MA}, pages = {2011--2028}, url = {https://www.usenix.org/conference/usenixsecurity22/presentation/chen-yunang-experimental}, publisher = {USENIX Association}, month = aug }
```

[Download](#)

[\[image: PDF icon\] Chen PDF](#)

[\[image: image\]](#)

[View the slides](#)

Presentation Video

Archived from <https://www.usenix.org/conference/usenixsecurity22/presentation/chen-yunang-experimental>. Rendered offline from the local Markdown copy.